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CPSC 321

# Homework 3:

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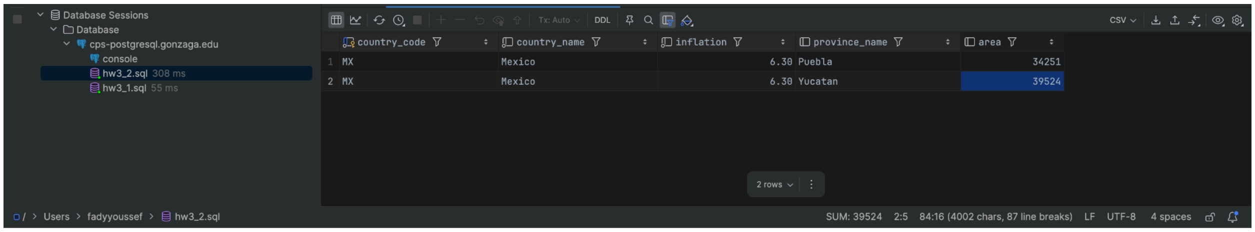
1. One challenge was writing queries that needed more than a simple filter. For example, finding provinces with two cities over 500k population was not obvious at first. Using COUNT with GROUP BY solved it and gave a clean answer. Another challenge was avoiding duplicate results when joining tables. Adding DISTINCT or making sure both province and country matched fixed those issues.

2. Each has four provinces, and each province has four cities with realistic populations. Borders were set only where they exist in real life. The tables are sized as follows:
   country (4 rows, 4 columns)
   province (16 rows, 3 columns)
   city (64 rows, 4 columns)
   border (4 rows, 3 columns)

   No extra rows were added for Part 2. The data from Part 1 already had enough variety to test queries without needing many changes.

3. Query: 1

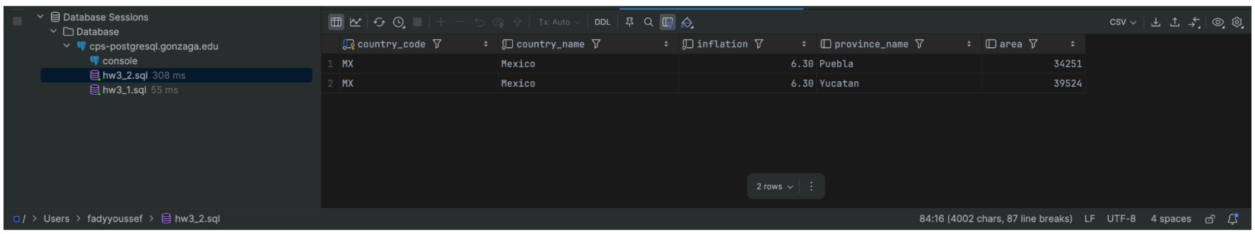


| | country_code | country_name | inflation | province_name | area |
|---|---|---|---|---|---|
| 1 | MX | Mexico | 6.30 | Puebla | 34251 |
| 2 | MX | Mexico | 6.30 | Yucatan | 39524 |

Example that does NOT match (small province in a low-inflation country):

INSERT INTO province VALUES ('Testshire','UK',30000);

Query: 2

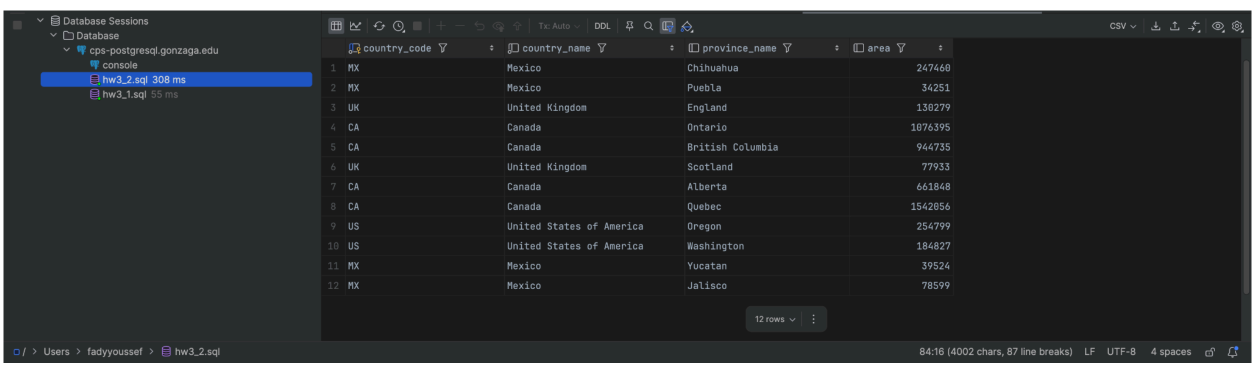


| | country_code | country_name | inflation | province_name | area |
|---|---|---|---|---|---|
| 1 | MX | Mexico | 6.30 | Puebla | 34251 |
| 2 | MX | Mexico | 6.30 | Yucatan | 39524 |

Example that does NOT match (same idea as Q1, JOIN version):

INSERT INTO province VALUES ('Minishire','UK',25000);

Query: 3

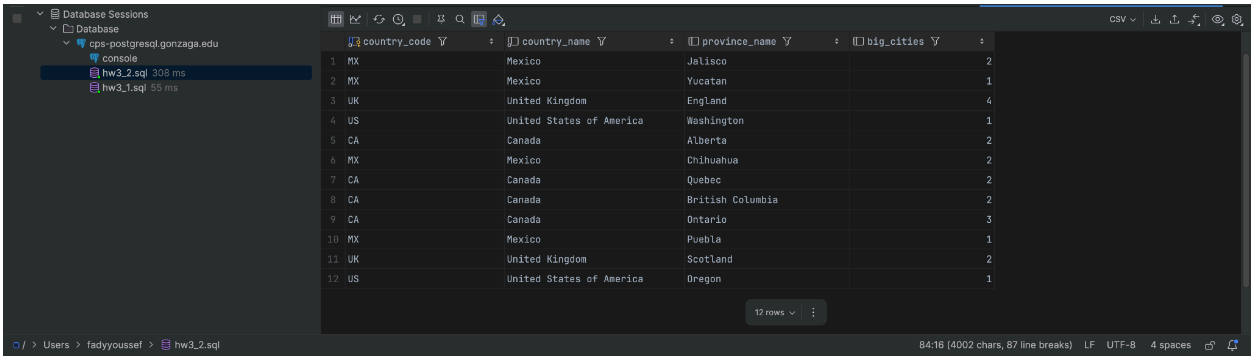


| | country_code | country_name | province_name | area |
|---|---|---|---|---|
| 1 | MX | Mexico | Chihuahua | 247460 |
| 2 | MX | Mexico | Puebla | 34251 |
| 3 | UK | United Kingdom | England | 130279 |
| 4 | CA | Canada | Ontario | 1076395 |
| 5 | CA | Canada | British Columbia | 944735 |
| 6 | UK | United Kingdom | Scotland | 77933 |
| 7 | CA | Canada | Alberta | 661848 |
| 8 | CA | Canada | Quebec | 1542056 |
| 9 | US | United States of America | Oregon | 254799 |
| 10 | US | United States of America | Washington | 184827 |
| 11 | MX | Mexico | Yucatan | 39524 |
| 12 | MX | Mexico | Jalisco | 78599 |

Example that does NOT match (province with only sub-500k cities):

INSERT INTO city VALUES ('Testville','Montana','US',490000);

Query: 4

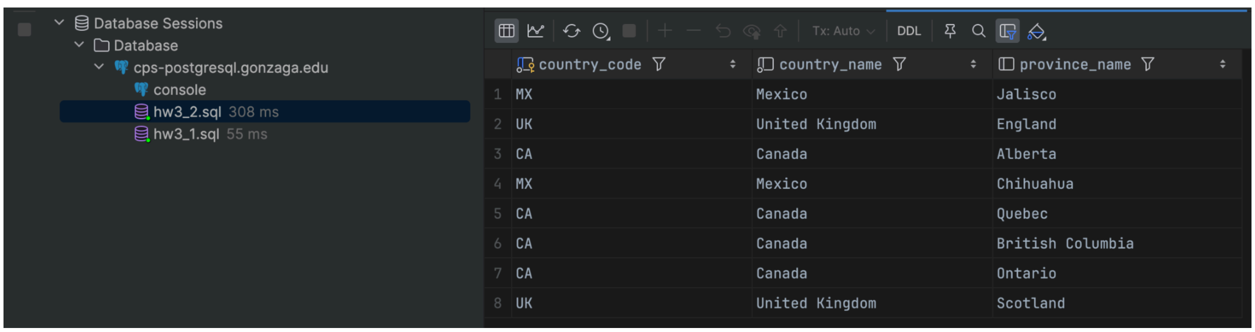



| | country_code | country_name | province_name | big_cities |
|---|---|---|---|---|
| 1 | MX | Mexico | Jalisco | 2 |
| 2 | MX | Mexico | Yucatan | 1 |
| 3 | UK | United Kingdom | England | 4 |
| 4 | US | United States of America | Washington | 1 |
| 5 | CA | Canada | Alberta | 2 |
| 6 | MX | Mexico | Chihuahua | 2 |
| 7 | CA | Canada | Quebec | 2 |
| 8 | CA | Canada | British Columbia | 2 |
| 9 | CA | Canada | Ontario | 3 |
| 10 | MX | Mexico | Puebla | 1 |
| 11 | UK | United Kingdom | Scotland | 2 |
| 12 | US | United States of America | Oregon | 1 |

Example that does NOT match (province with zero big cities):

INSERT INTO city VALUES ('Smalltown','Maine','US',120000);

Query: 5

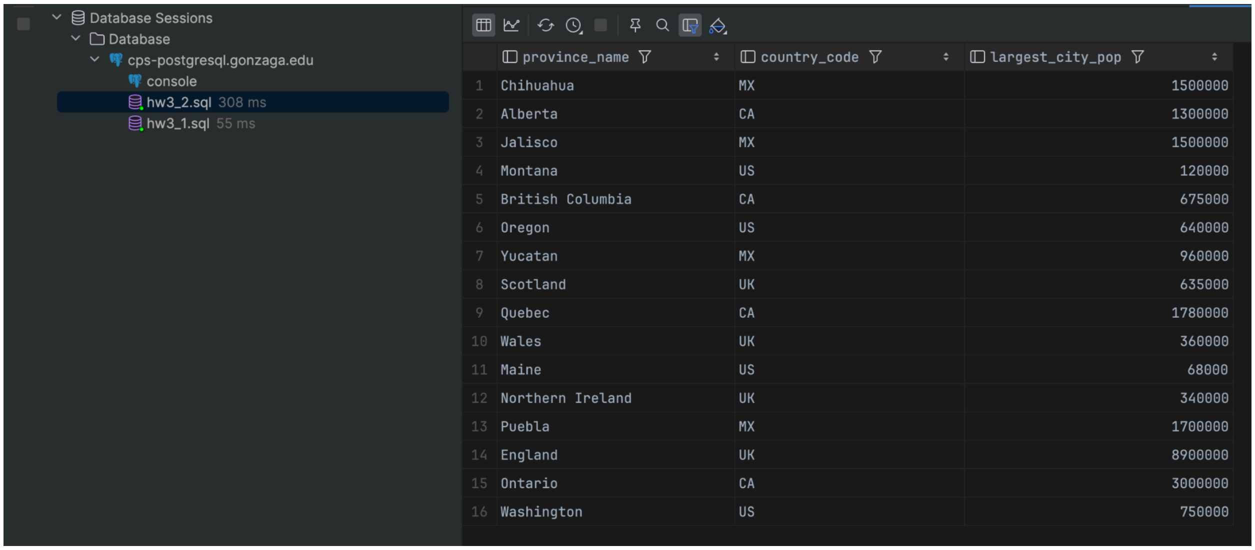



| | country_code | country_name | province_name |
|---|---|---|---|
| 1 | MX | Mexico | Jalisco |
| 2 | UK | United Kingdom | England |
| 3 | CA | Canada | Alberta |
| 4 | MX | Mexico | Chihuahua |
| 5 | CA | Canada | Quebec |
| 6 | CA | Canada | British Columbia |
| 7 | CA | Canada | Ontario |
| 8 | UK | United Kingdom | Scotland |

Example that does NOT match (province with exactly one big city, still < 2):

INSERT INTO city VALUES ('AlmostThere','Washington','US',400000);

Query: 6

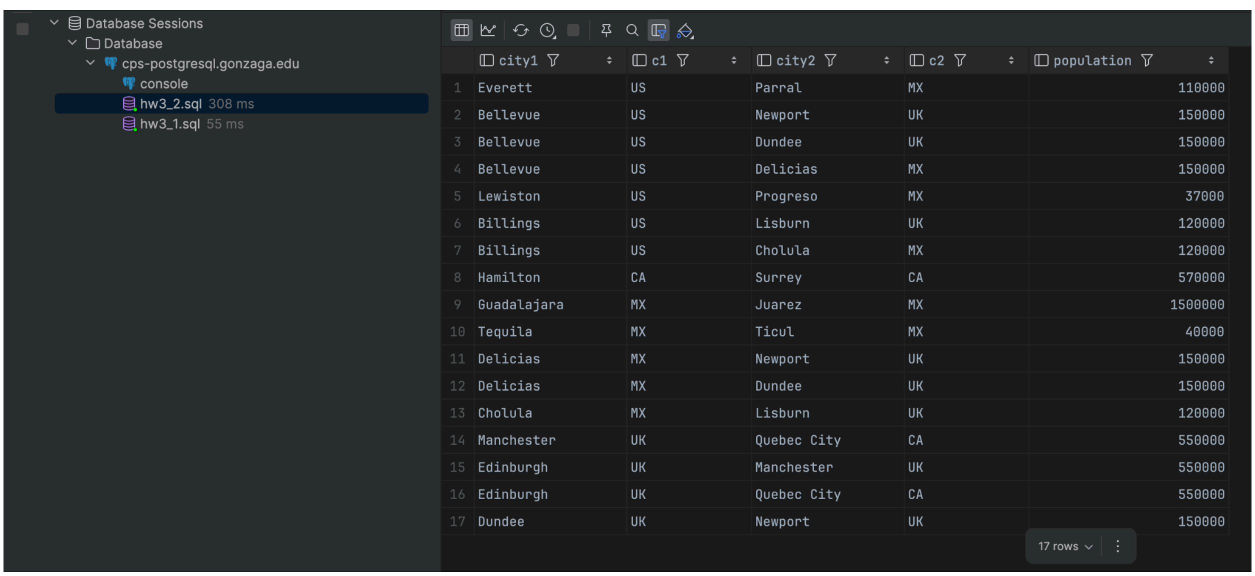



| | province_name | country_code | largest_city_pop |
|---|---|---|---|
| 1 | Chihuahua | MX | 1500000 |
| 2 | Alberta | CA | 1300000 |
| 3 | Jalisco | MX | 1500000 |
| 4 | Montana | US | 120000 |
| 5 | British Columbia | CA | 675000 |
| 6 | Oregon | US | 640000 |
| 7 | Yucatan | MX | 960000 |
| 8 | Scotland | UK | 635000 |
| 9 | Quebec | CA | 1780000 |
| 10 | Wales | UK | 360000 |
| 11 | Maine | US | 68000 |
| 12 | Northern Ireland | UK | 340000 |
| 13 | Puebla | MX | 1700000 |
| 14 | England | UK | 8900000 |
| 15 | Ontario | CA | 3000000 |
| 16 | Washington | US | 750000 |

Example that does NOT show up as the "max" (adds a smaller city to the province):

INSERT INTO city VALUES ('NotMaxCity','Quebec','CA',12345);

Query: 7

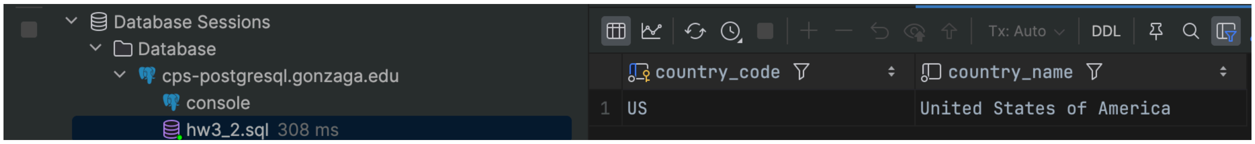



| | city1 | c1 | city2 | c2 | population |
|---|---|---|---|---|---|
| 1 | Everett | US | Parral | MX | 110000 |
| 2 | Bellevue | US | Newport | UK | 150000 |
| 3 | Bellevue | US | Dundee | UK | 150000 |
| 4 | Bellevue | US | Delicias | MX | 150000 |
| 5 | Lewiston | US | Progreso | MX | 37000 |
| 6 | Billings | US | Lisburn | UK | 120000 |
| 7 | Billings | US | Cholula | MX | 120000 |
| 8 | Hamilton | CA | Surrey | CA | 570000 |
| 9 | Guadalajara | MX | Juarez | MX | 1500000 |
| 10 | Tequila | MX | Ticul | MX | 40000 |
| 11 | Delicias | MX | Newport | UK | 150000 |
| 12 | Delicias | MX | Dundee | UK | 150000 |
| 13 | Cholula | MX | Lisburn | UK | 120000 |
| 14 | Manchester | UK | Quebec City | CA | 550000 |
| 15 | Edinburgh | UK | Manchester | UK | 550000 |
| 16 | Edinburgh | UK | Quebec City | CA | 550000 |
| 17 | Dundee | UK | Newport | UK | 150000 |

Example that does NOT form a pair (unique population):

INSERT INTO city VALUES ('Uniqville','Oregon','US',1234567);

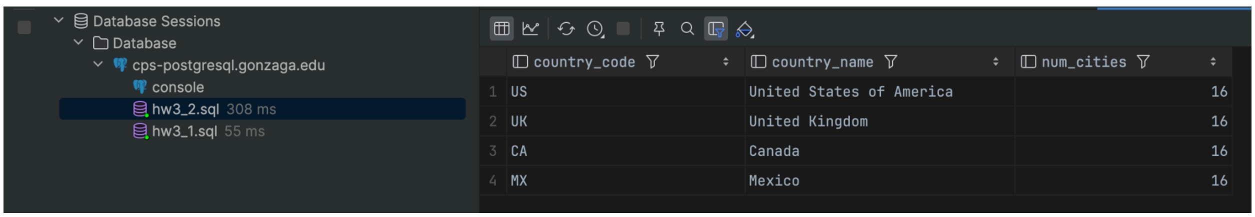
Query: 8




Example border that does NOT satisfy the high/low condition (still valid row):

INSERT INTO border VALUES ('CA','US',8891.00);

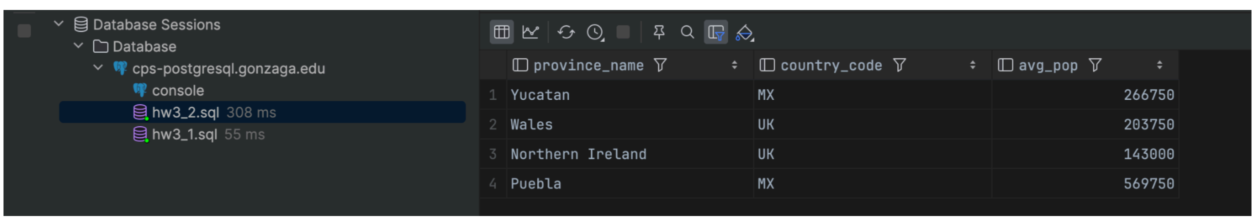
Query: 9




Example country that does NOT appear (no cities for this country in dataset):

INSERT INTO country VALUES ('ES','Spain',42000,3.00);

Query: 10




Example province that does NOT match (area too large):

INSERT INTO province VALUES ('Bigshire','MX',50000);